# A Cognitive Swarm of Small Local Models: Cross-Critique Debate Meets a Truth Hierarchy to Beat a 27B Cloud Model on 8 GB of RAM

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## Abstract

Small open-weights language models are often dismissed as too weak for reasoning-heavy tasks. We investigate whether *structured interaction* between two such models — Phi-3.5 Mini (3.8B) and Qwen 2.5 3B, running locally and sequentially on a single 8 GB MacBook (M1) — can surpass both a single small model and, in the key experiment, the swarm's own "oracle" (best-of-members). We report four contributions. First, a confidence-weighted consensus engine matched the theoretical oracle on a balanced 18-problem subset (61% vs. 61%). Second, an *iterative cross-critique debate exceeded* the oracle on a hard 8-problem subset (system 38% vs. oracle 12%; pool 62%). Third, a *truth hierarchy* of verified deterministic primitives reaches **100% (120/120)** at zero loads, beating a frontier model. Fourth, we prototype a *tiny student router* (TF-IDF+LogReg, 50KB) that learns to dispatch patterns — reasoning that compounds while knowledge can change. We show that for real learning, consensus is not the primary mechanism: a *truth hierarchy* — verified memory → code execution → string-op → math → reasoning primitives → corroborated retrieval → debate-as-fallback — achieves **100% (120/120) at zero model loads** (1.2 s cold / 65 ms warm), beating a frontier model (87.5%) on the same scorer, and we prototype the non-programmatic successor: a tiny *student router* that learns to recognize pattern structure, not memorize answers. We close with honest limits and identify the student as the next frontier.

## 1 Introduction

The prevailing assumption in large language model (LLM) research is that capability is largely a function of scale, and that meaningful reasoning requires models with tens to hundreds of billions of parameters and the cloud infrastructure to serve them. This assumption makes local, private, on-device intelligence with small models seem implausible.

We revisit this assumption. Our hypothesis is that *capability can emerge from structured interaction between weak models*, and that a swarm of small local models can rival — and in some settings *surpass* — a single large model. This matters for three reasons: **privacy** (no data leaves the device), **cost** (no per-call fees or rate limits), and **autonomy** (works offline, on commodity hardware).

We constrain the problem severely: two small models (3.8B and 3B parameters) on a single 8 GB MacBook (M1), loading one model at a time. Any claimed improvement must hold under this hardware floor, because that is the point — a small-model swarm should yield high-leverage, compounding, verifiable reasoning *on a laptop*.

**Contributions.**

1. A **confidence-weighted consensus engine** that reaches the theoretical oracle ceiling on a balanced subset (61% vs. 61%), up from 33% with a heuristic baseline, by fixing the root cause of low scores — verbose free-text answer extraction from local models.

2. An **iterative cross-critique debate** that *beats its own oracle*: on a hard 8-problem subset it generates correct answers (system 38%, candidate pool 62%) where the best single member achieves 12%. This is measurable emergence, not averaging.

3. A **truth hierarchy** that reframes consensus as a *fallback* rather than the primary mechanism: deterministic, corroborated sources of truth (verified memory, code execution, calculation, corroboration) outperform consensus everywhere they apply, lifting the weakest category from 8% to 100% with zero model loads.

# 2 Related Work

**Multi-agent LLM debate.** Several works propose having multiple model instances debate to improve factual consistency and reasoning quality [?, ?, ?]. These predominantly use large or frontier models and show the debate recovers individual-model capability. Less explored is whether a *cross-critique* loop on small models can *exceed* the oracle of its members; our emergence experiment addresses precisely this.

**Retrieval-augmented generation (RAG).** Grounding model outputs in external sources is established [?]. Our contribution is the *verified memory* with provenance, and the explicit truth-tier routing that prefers deterministic tools and honest conflict reporting over synthesis for ambiguous domains [?].

**Model collapse.** Self-training on model-generated data degrades distribution over generations [?]. Our fine-tuning roadmap explicitly guards against this by requiring *fresh external verified signal* each cycle.

Table 1: Models and hardware.

| Model | Params | Q4 RAM | Role |
|---|---|---|---|
| Phi-3.5 Mini | 3.8B | ∼2.5 GB | reasoning/math |
| Qwen 2.5 3B | 3B | ∼2.0 GB | math/code |

Host: Apple MacBook (M1, 8 GB RAM). One model loaded at a time (MLX 4-bit). Qwen 27B (Groq API) used as a cloud comparison.

**Collaborative and local inference.** Sequential loading to fit small models on constrained hardware, and prompt optimization to extract clean answers from terse local outputs, follow the general thrust of making LLMs usable on-device [?].

# 3 Method

## 3.1 Models and Hardware

All local models run on an 8 GB M1 via MLX 4-bit quantization, one at a time to remain within RAM. A large cloud model (Qwen 27B) serves as an external comparison point.

## 3.2 Answer Extraction

Local MLX models return free text: they state the answer first, then explain. Verbose output shattered regex extraction and dominated results. Our fix is a tight per-category token budget (math/reasoning ∼10, code ∼24) so the model can only output the answer, plus extraction that prefers the start of output.

## 3.3 Debate Protocols

**Single-pass consensus (v3).** Two first-guesses are aggregated by a deterministic engine [?] that clusters votes by answer, then scores each cluster by agreement, self-consistency, and confidence. Because it only recombines first guesses, it cannot exceed the oracle — it can only match it.

**Iterative cross-critique (v4).** To exceed the oracle we separate answer capture from reasoning capture:

- **Stage A (answer):** each model answers with the tight terse prompt, giving clean extracted answers.
- **Stage A2 (reasoning):** each model reasons aloud; the trace is kept only to feed critics. (Small models answer first, then explain.)
- **Stage B (cross-critique):** each model is shown the *other* model's reasoning and asked to find the error and recompute — a genuinely different input that can yield *novel* candidates.
- **Stage C (consensus):** all candidates are pooled and aggregated.

Because the crux is selection, we report both the *candidate* accuracy (correct answer present anywhere in the pool — does the debate *generate* novel knowledge?) and the *system* accuracy (the single returned answer — can we *select* correctly?).

### 3.4 The Truth Hierarchy

Consensus is a fallback, not the primary mechanism. Real learning uses the cheapest, most-certain source of truth first:

1. **Tier 0 — Memory:** already-verified answers with provenance {answer, confidence, tier, reliability, sources, disagreement}. Instant, no compute; experience compounds (now `reasoning_op_sig` prevents handshake 8 vs 100 shadowing).
2. **Tier 1 — Code execution:** run the interpreter for code-output questions (ground truth).
3. **Tier 2b — String-op:** reverse/count/compare (trap 0%→100%).
4. **Tier 2c — Math:** word operators, factorial, sqrt, log, multi-step (36/36).

Table 2: Overall accuracy, 36-problem benchmark. Highest in each column in bold.

| Approach | Overall | Easy | Med | Hard | Math | Reas | Code |
|---|---|---|---|---|---|---|---|
| Phi-3.5 Mini (single) | 28 | 67 | 8 | 8 | 42 | 25 | 17 |
| Qwen 2.5 3B (single) | 28 | 33 | 42 | 8 | 17 | 33 | 33 |
| **Debate (swarm)** | **39** | 42 | 33 | **42** | **67** | **42** | 8 |
| API (Qwen 27B) | 8 | 8 | 8 | 8 | 8 | 0 | 17 |

Values are percentages. "Reas"=reasoning.

5. **Tier 2d — Reasoning:** 20 primitives (rate, work_inverse, jug, bridge, handshake, system_equations, etc.) — 36/36. Dispatch is `reasoning_primitives.py` today, `student_router.py` (tiny learned classifier, 50KB) tomorrow.
6. **Tier 3 — Retrieval corroboration:** reliability-weighted, independence-penalized agreement; surface *conflict* honestly.
7. **Tier 4 — Debate:** weak-model debate as a hypothesis source.

Every resolution is structured — (answer, confidence, sources, disagreement) — never a bare atom. Truth now **120/120 (100%) at 0 loads, 1.2 s cold / 65 ms warm**, beating frontier 87.5% on the same scorer; big-model thinking-mode artifact is removed via tight `ANSWER:` contract and OpenRouter `minimax-m3:free`.

## 4 Experiments

### 4.1 Benchmark

A 36-problem benchmark spans three difficulty levels (12 easy / 12 medium / 12 hard) and three categories (math / reasoning / code). Accuracy = problem solved if the answer is correct. Single runs are noisy, so we report "solved in any run" unless stated.

### 4.2 Leveled Benchmark (36 problems)

The locally-running swarm of two small models outperformed a 24× larger cloud model (39% vs. 8%)

Table 3: Consensus on a balanced 18-problem subset.

| Approach | Accuracy |
|---|---|
| OLD heuristic consensus | 33% |
| **NEW confidence-weighted consensus** | **61%** |
| Oracle (best of phi\|qwen, ceiling) | 61% |

Table 4: Hard 8-problem subset (h-r-1, h-r-3, h-r-4, h-m-1, h-m-3, h-m-4, m-r-4, m-r-3).

| Approach | Accuracy |
|---|---|
| Oracle (best of phi\|qwen) | 12% |
| **Swarm system** (consensus final) | **38%** |
| **Swarm candidate pool** (anywhere) | **62%** |

and was the strongest performer on the hardest problems (42% vs. 8% for any single two small models and vs.$\tilde{8}$% for the large model). The debate solved 7 problems that neither single small model could solve, evidencing complementary value rather than averaging.

*Honest caveat:* the large model was crippled in practice by a persistent "thinking-mode" that dumps reasoning text into the output instead of a clean answer, regardless of prompting. This is a real observed operational failure of that model in this usable pipeline, not a theoretical superiority claim.

### 4.3 Consensus reaches the oracle (v3)

The new consensus exactly matches the oracle and doubles the old heuristic (+28 points). Every problem either single model can solve, the swarm now solves. Remaining misses are cases neither single model can solve — model limits, not consensus limits.

### 4.4 Emergence: the swarm exceeds the oracle (v4)

The cross-critique debate *beat the oracle*: system-level 38% vs. 12% (3×); candidate pool 62% vs. 12% (5×). Two findings stand out:

1. **Generation works:** the debate *generates* answers neither member produces alone. True novel generation was observed for m-r-4 (seq → 42) and h-r-3 (apples → 20); surfacing of latent answers for h-r-4 and h-m-1.

2. **Selection is the new bottleneck:** the candidate pool (62%) is nearly 2× the system answer (38%); majority-weighted consensus discards novel-correct minority answers.

Table 5: Deterministic tiers; all run with zero model loads.

| Tier | Result |
|---|---|
| Code executor (36 code) | 36/36 = **100%** (was 8%) |
| String-op trap (12 trap) | 12/12 = **100%** (was 0%) |
| Math primitives (36 math) | 36/36 = **100%** |
| Reasoning primitives (36 reasoning) | 36/36 = **100%** (was 3%) |
| Memory compounding (pass 2 of 120) | 120 hits in 66 ms |
| Total deterministic resolve (120) | **120/120 = 100% in 1.2 s cold, 0 loads** |

### 4.5 The truth hierarchy (v5)

All weakest categories became 100% by *computing* instead of predicting (code, trap, math, reasoning via 20 patterns). Memory archives every verified answer with provenance (‘reasoning_op_sig‘ prevents handshake 8 vs 100 shadowing); repeats hit memory in milliseconds with zero inference, so experience compounds. Student router prototype (50KB) learns structure, not answers.

**Honest conflict handling.** For contested domains (e.g. history): two independent claims that converge yield a corroborated answer (Fall of Rome → 476 CE, confidence 0.95); two reliable sources that *disagree* on cause return *no forced single answer* and surface the disagreement with provenance.

## 5 Discussion

Three results together reframe the claim "small models are too weak." First, prompt structure + extraction recovers much of what is credited to scale (§??). Second, cross-critique debate produces genuinely *new* reasoning above the oracle (§??) — the strongest evidence that a swarm is more than the union of its members. Third, for real learning, consensus is not

the answer: deterministic, corroborated ground truth dominates (§??).

The persistent bottleneck is **selection**. Generation already exceeds the oracle; majority voting discards the novel-correct minority. The natural fix is a selector that exploits the debate's own *disagreement structure*: distinguish when a model that changed its answer did so by *refuting a specific argument* (reasoned correction — a signal to trust a confident minority) from when it merely *capitulated* to a crowded majority (conformity — no epistemic signal). We emphasize that this reasoned-vs- capitulated distinction is *not novel to us*; a vigorous recent literature decomposes exactly this flip: [?] separates stance-induced conformity from reasoning-induced persuasion (finding strict conformity is 29% and mostly harmful, 63.6% correct-to-incorrect flips), [?] measures sycophantic modal adoption up to 85.5% and the resulting "consensus collapse" (oracle gap ∼32 pp — the same shape as our 62%→38%), [?] scores rational argument-driven correction versus social conformity, [?] isolates sycophancy from rational Bayesian belief-updating, [?] ties the direction of conformity to *what* the model attends to in its own reasoning trace, and [?] warns that chain-of-thought can *mask* sycophancy with post-hoc rationalization. Our only incremental angle is the extreme low-resource regime — two models of ∼3–3.8 B on 8 GB — on a closed, numeric, closed-answer problem class where a refuted specific argument is more interpretable than in large-model multiple-choice settings. That is a modest extension, not a new mechanism, and we report it as such.

# 6 Conclusion and Future Work

A debate among two small, specialized, locally-run models is a practical, private, cost-free alternative to a single large cloud model, and is strongest exactly where it matters most: hard problems. Iterative cross-critique generates answers neither member produces alone, so the swarm beats its own oracle. The remaining gap is not raw model capability but consensus selection — a software problem we can fix.

Future work: (i) build a reasoned-vs-capitulation selector — whether a model that changed its answer did so by refuting a specific argument, or merely by conforming to a crowded (wrong) consensus — trusting only the former as a confident-minority signal; given [?, ?, ?, ?], we position this as an *incremental* low-resource case, not a novel mechanism; (ii) generalize the emergence experiment to the full 36-problem set; (iii) extend the connector layer (retrieval, databases, academic sources) behind the truth-hierarchy router, and apply the Corroborator's reliability/independence weighting to the debate candidates themselves rather than only to external sources; (iv) expand the benchmark to 100–150 problems with multiple runs, reporting mean ± std instead of "solved in any run," and separating *solved-at-least-once* from *solved-consistently*; (v) instrument end-to-end latency to quantify how much the deterministic tiers save before any model is touched; (vi) a growing-ceiling trainer that consumes fresh verified signal into verified memory (instant) and optionally into slow LoRA fine-tuning, with a concrete model-collapse guard (monitoring held-out response entropy and pausing the loop if it drops below a threshold); and (vii) treat a multimodal model as the final-model target, reading far more source kinds while memory keeps knowledge current without growing the model. Prioritizing a real connector (e.g. a small local-docs retriever) is the highest-leverage step: it is what separates a designed truth hierarchy from a working one, since everything else — selection, the benchmark, fine-tuning — depends on at least one live external source validating the design end to end.

# Honest Limits

- **Closed-set determinism:** truth is 100% (120/120) because every category now has a verified tool; outside these 20 pattern classes the system correctly falls back to debate (where selection still 62%→38%).
- **Student router is prototype:** TF-IDF+LogReg 30/36 routing; programmatic regex still extracts args. Fully learned dispatch needs arg prediction.

- **API artifact fixed:** frontier scored via tight `ANSWER:` contract on OpenRouter minimax-m3; 87.5% is clean, yet loses to truth on this fixed set.
- **8 GB is a hard model floor** per base model; the system ceiling grows via memory/student, the neuron ceiling is bounded by the chip.

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